

Le Anh Luong

BACKEND DEVELOPER (.NET) · AI / LLM

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SUMMARY

Backend developer with 2.5 years of .NET experience building microservices for a multi-tenant SaaS platform in an 8-person backend team. Own the omni-channel chat and telephony backend end-to-end, lead development on marketing & notification and HRM; plus payment & e-invoicing integrations — and design the shared-kernel patterns the platform's services reuse for consistency. Architected and operate a self-hosted Generative AI stack: a RAG pipeline running LLM + embeddings on our own GPU — customer data never leaves the company, with no cloud-API cost.

TECHNICAL SKILLS

- **Languages & Backend:** C#, SQL, .NET 10 / ASP.NET Core, EF Core, gRPC, SignalR
- **Architecture:** Microservices, Clean Architecture, DDD, CQRS (MediatR), multi-tenant RLS, API Gateway (YARP)
- **Messaging & async:** RabbitMQ / MassTransit, event-driven (outbox, idempotency keys), Hangfire (background & scheduled jobs), async webhook processing
- **AI / LLM:** RAG (pgvector + embedding + LLM), self-hosted inference (vLLM + Qwen2.5-7B, HF TEI + bge-m3), OpenAI-compatible abstraction (self-host ↔ cloud by config), function calling, multi-turn context, intent classification, speech-to-text, MCP (Model Context Protocol)
- **Data & infrastructure:** PostgreSQL 16 (JSONB, pgvector), Redis 7, Meilisearch, Docker, MinIO, Serilog + OpenTelemetry, CI/CD
- **Integrations:** VNPay / VNPayQR / VietQR, BKA V & Viettel (e-invoice & e-contract), Brevo (email/SMS), Zalo ZNS, Facebook Messenger, Instagram, Zalo OA, Hikvision / ZKTECO (ISAPI), FreePBX / Asterisk
- **Security:** JWT RS256 / JWKS, OAuth2, Argon2id, AES-256-GCM, 2FA TOTP, RBAC, rate limiting
- **Testing:** unit & integration testing (xUnit) · **Other:** TypeScript, Next.js / React, Node.js, Electron

EXPERIENCE

Backend Developer (.NET)

NextX Future JSC 04/2024 – Present · Hanoi

Multi-tenant SaaS business-management platform, built by an 8-person backend team. Started with feature development & maintenance on the live system, then rebuilt the platform into a .NET 10 microservice architecture on a shared kernel.

Since 11/2025 · Platform rebuild into microservices (.NET 10)

Ported business rules out of the live legacy PHP system into a Clean Architecture design on the existing data schema; built microservices on an internal NuGet shared kernel.

AI Chatbot (RAG) — designed and operated end-to-end

- Built the RAG engine: embed the business knowledge base → top-K vector search on pgvector → LLM rephrases strictly from retrieved context to prevent hallucination. Added multi-turn context, intent classification, function calling against live data, and automatic escalation to a human agent below a confidence threshold.
- Stood up the in-house GPU inference stack: Qwen2.5-7B on vLLM + bge-m3 embeddings on HF TEI, both exposing OpenAI-compatible endpoints so local ↔ cloud is a config switch. Migrated the entire chatbot off OpenAI cloud onto self-hosting, re-embedding the whole knowledge base (1536 → 1024 dimensions) — customer data never leaves company infrastructure and the cloud-API cost is gone.

MCP Server — opening the platform to AI assistants

- Built an MCP (Model Context Protocol) server in TypeScript/Node so Claude, GPT and Gemini can call NextX APIs directly — around 45 tools across CRM, Sales, Comm, Office, Gym, Golf, HRM and DMS.
- OAuth2 client_credentials authentication (agent JWT) with token caching and refresh; every call is routed through the API Gateway and constrained by per-route scopes, so an AI assistant cannot exceed its permissions.

Comm — Omni-channel Chat · owner

- Unified inbox across 4 channels: Facebook Messenger, Instagram, Zalo OA and Zalo Personal (custom Node adapter); real-time over SignalR.
- Conversation distribution via smooth weighted round-robin (NGINX algorithm): weight / skill / priority strategies, presence-aware, Redis distributed lock to prevent races between concurrent webhooks.
- Webhooks return 200 OK immediately and process asynchronously through Hangfire; channel credentials encrypted with AES-256-GCM; contact/lead sync to CRM & Sales over gRPC.

CRM & Telephony (Call) — core contributor

- Contributed to the CRM service: Lead / Contact / Opportunity / Quotation plus analytics, exposed over gRPC; built lead auto-assignment (rule-based routing + least-load or weighted round-robin selection with distributed locking) and lead scoring into Hot / Warm / Cold tiers.

- Owned the telephony backend, integrated with FreePBX / Asterisk (VoIP infrastructure operated by another team): call-log and recording sync, attaching calls to CRM contacts/leads, and speech-to-text transcription and transcript analysis of recordings for storage and search.

Fintech — Payment, e-invoice & e-contract integrations

- Online payments via VNPay / VNPayQR / VietQR (QR generation, idempotent IPN handling, reconciliation); e-invoice and e-contract integration with BKAV, Viettel and MISA — digital signing, issuance, lookup and status sync.

Marketing & Notification · lead developer

- Bulk campaign engine for Email/SMS (Brevo) and Zalo ZNS: MassTransit workflows, Hangfire batch dispatch, MJML templates, sender verification with DKIM/SPF; multi-channel Notification service (in-app / push / email) exposed over gRPC to the whole platform.

HRM & Attendance · developer

- HRM: attendance, leave, payroll & social insurance, labor contracts, reporting and export.
- Built hik-agent (Electron) bridging Hikvision / ZKTeco attendance devices to the server over ISAPI + SignalR: card and face enrollment sync, real-time check-in events, remote door unlock.

SharedKernel — cross-cutting framework reused across services (internal NuGet)

- Import and Export/Report frameworks: each service declares its own columns and duplicate-match keys through a registry + descriptors; inline execution for small files, queued for large ones; standardized job status and scheduled exports.
- Event-driven notifications on RabbitMQ: designed the **NotificationRequested** thin-trigger contract — producers send only type + data + recipient while Notification resolves channels and renders per locale; idempotency keys against duplicate processing, outbox pattern. Migrated every service's producers onto this contract.
- Search framework (Meilisearch): platform-wide ISearchClient abstraction — swap engines without touching callers, with a NullSearchClient so search degrades instead of failing the request. Tenant isolation enforced in the client: workspaceId is required and ANDed into every query so a caller cannot leak across tenants (from a security-audit finding); standard {service}-{entity} index naming, event-driven sync on entity change plus Hangfire backfill/delta jobs.
- Worked on the YARP API Gateway and IAM: JWT RS256 + JWKS, Argon2id, 2FA TOTP, RBAC, rate limiting; multi-tenant isolation via PostgreSQL Row-Level Security + EF global query filters.

04/2024 – 10/2025 · Earlier — feature development & maintenance on live products (ASP.NET Core)

- GymMaster — gym & sports-club management (NextX in-house product), primary work: membership module (packages, court/lane booking, session deduction), the promotions engine, member check-in with check-in history, and staff attendance/timekeeping.
- RMIT — room & court booking system for students: single sign-on login and schedule-based room/court reservation.
- Pickleball & golf court booking: slot/schedule-based booking with conflict handling, membership and play packages across venues.
- Telephony (Call) backend: built the call backend on top of a ready-made third-party VOIP API (infrastructure operated by another team) — call-log handling and attaching calls to CRM records.
- CRM: built reports and maintained/bug-fixed the live CRM modules.

Process: since 11/2025 working fullstack (.NET backend + Next.js/React frontend) with AI coding assistants (Claude Code, Codex) — quality held by code review, unit/integration tests and shared-kernel architectural constraints.

.NET Developer Intern

FPT Software 05/2023 – 09/2023 · Hanoi

- Joined a .NET web development team: built REST APIs with ASP.NET Core + EF Core on SQL Server, wrote unit tests and API documentation; worked in Scrum — Jira tasks, pull requests and code review, fixed bugs from QA tickets.

PERSONAL PROJECTS

AA Catering Management System — graduation project

- Enterprise catering-management system: Next.js + ASP.NET Core, real-time updates over SignalR, Excel report export, SQL Server, packaged with Docker.

EDUCATION & LANGUAGES

FPT University (Hoa Lac) — B.E. Software Engineering · Graduated Dec 2024

English: basic — able to read technical documentation